Does Pre-training Induce Systematic Inference?
How Masked Language Models Acquire Commonsense Knowledge

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Overview

Test simple interventions on MLM pre-training data

Goal: Understand how MLMs acquire commonsense knowledge

Method:

- swap verbalized knowledge into pre-training minibatches of BERT
- evaluate generalization to supported inferences after training on these minibatches

Takeaway

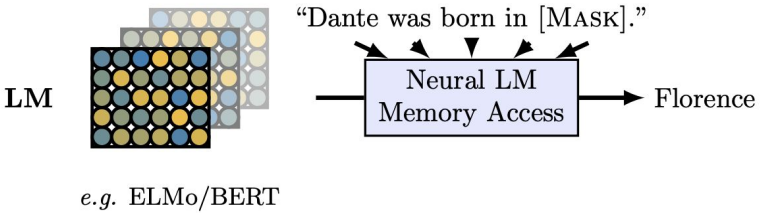
Generalization of MLMs to supported inferences:

- peaks early in pre-training
- sometimes degrades during the latter half of pre-training

Which suggests: knowledge is not systematically acquired by the inferences studied.

Motivation

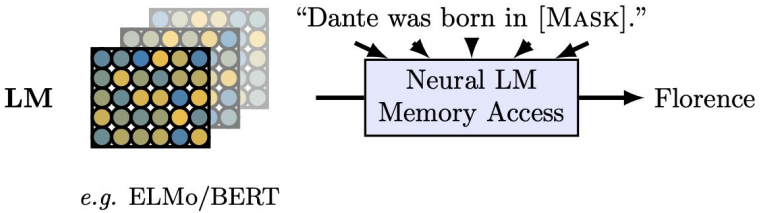
MLMs encode commonsense knowledge.



(Figure reproduced from Petroni et al., 2019)

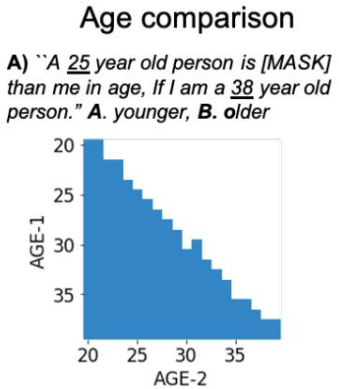
Motivation

MLMs encode commonsense knowledge.



(Figure reproduced from Petroni et al., 2019)

Sometimes in seemingly systematic ways.



(Figure reproduced from Talmor et al., 2020a)

Q: Do MLMs acquire knowledge via *systematic inference?*

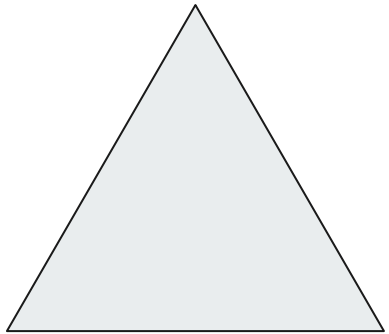
E.g., learning **“robins can fly”** from:

1. **“robins are birds”**
2. **“birds can fly”**
3. **downward entailment** (A *is-a* B $\wedge$ B *has-property* C $\models$ A *has-property* C).

Systematic inference: the ability to learn general rules and apply them in novel settings, as opposed to learning only particular instances of the rule (Fodor and Pylyshyn, 1988)

Terminology

Super-statement: "A bird has a [MASK]." → wing



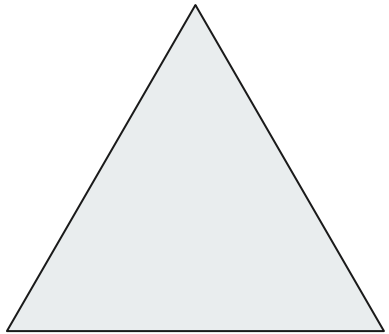
Sub-statement:
"A robin has a [MASK]." → wing

Class relation:
"A robin is a [MASK]." → bird

Terminology

*controls

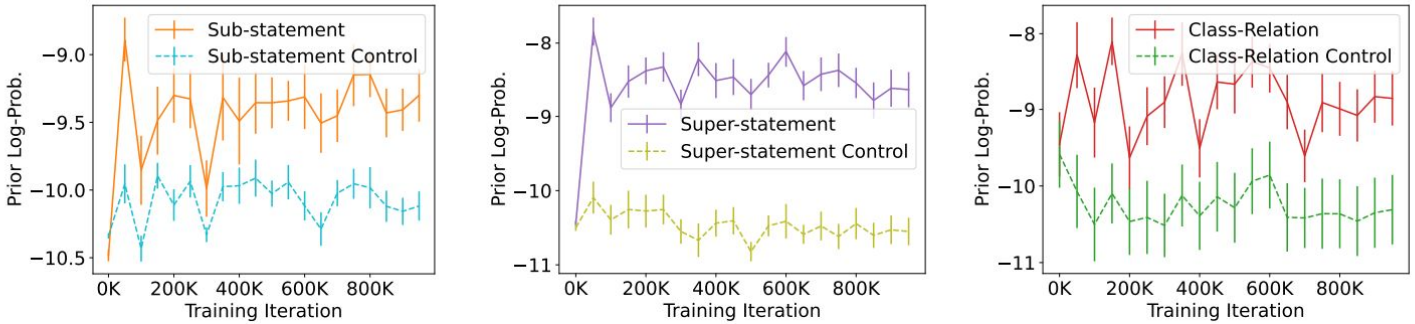
Super-statement: "A bird has a [MASK]." → wing / fin



Sub-statement:
"A robin has a [MASK]." → wing / fin

Class relation:
"A robin is a [MASK]." → fish

Results: Priors



(a) Sub-statement (b) Super-statement (c) Class Relation

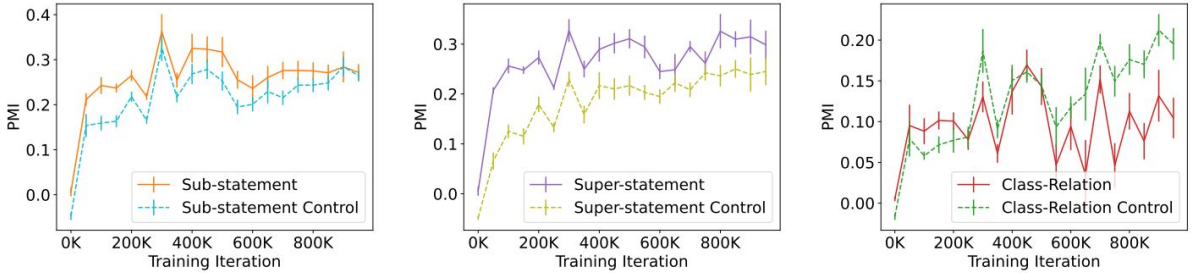
Figure 1: The prior log-probability of each knowledge type estimated by BERT across pre-training iterations.

Method

1. Intervene on BERT pre-training data.
2. After updating on intervening data, evaluate how predictions change.

E.g., swap "A robin can [MASK]." into a minibatch. After model updates, does knowledge "A robin is a bird." improve?

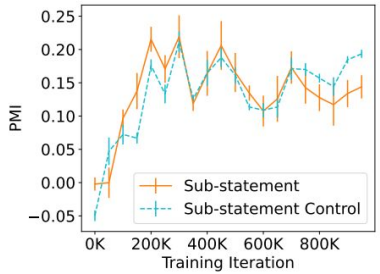
Results: Interventions



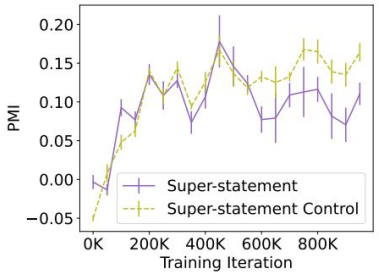
(a) Super-statement→Sub-statement (b) Super-statement→Super-statement (c) Super-statement→Class relation

(A -> B: inject A into pre-training data, then evaluate change in B)

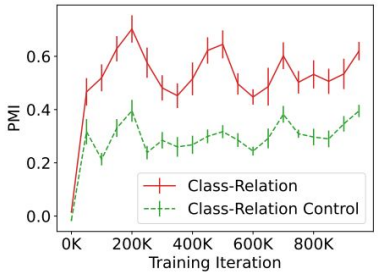
Results: Interventions



(d) Class relation→Sub-statement



(e) Class relation→Super-statement



(f) Class relation→Class relation

(A -> B: inject A into pre-training data, then evaluate change in B)

Conclusion

Generalization does not improve during pre-training.

This suggest knowledge studied is not acquired by these inferences.

Future improvements might consider enforcing such ability.